

# **Adaptive Fine-Tuning for Cross-Lingual Alignment via Layer-Wise Geometry**

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## **Motivation**

Multilingual models like XLM-R exhibit strong cross-lingual alignment, where semantically similar sentences map to nearby regions in embedding space. This enables transfer learning, cross-lingual retrieval, and multilingual representations. However, improving alignment through fine-tuning remains challenging.

While contrastive fine-tuning can improve alignment, prior methods typically apply static or heuristic updates across layers and ignore layer-wise differences. This overlooks an important structural property: alignment is not uniformly distributed across the network. Different layers capture different linguistic features and vary in alignment quality, so naive fine-tuning can introduce representation drift that degrades pretrained structure and harms other languages.

This project addresses this limitation by introducing a fine-tuning framework that adapts updates based on layer-wise alignment geometry, improving alignment where needed while preserving useful pretrained structure.

## **Proposed Research**

We propose an adaptive fine-tuning framework that scales updates at each layer based on alignment quality. First, we compute layer-wise alignment scores using parallel sentence pairs. For each layer, we extract sentence representations and measure cross-lingual alignment using metrics such as Procrustes distance or cosine similarity. Second, we introduce a regularization term that penalizes deviation from pretrained representations to prevent catastrophic drift and preserve global structure. Third, we use these scores to scale updates during fine-tuning: well-aligned layers receive smaller updates, while poorly aligned layers receive larger ones. Alignment scores are recomputed periodically so update scaling adapts as representations evolve.

## **Related Work**

Multilingual models such as XLM-RoBERTa learn partially aligned cross-lingual representations (Conneau et al., 2020). Prior work improves alignment via contrastive fine-tuning with objectives such as InfoNCE, bringing translation pairs closer while separating non-parallel ones. Methods such as AlignFreeze (Bakos et al., 2025) show that selectively freezing layers helps preserve pretrained structure, while discriminative fine-tuning (Howard and Ruder, 2018) assigns different learning rates across layers, typically based on depth.

Our approach builds on these ideas by introducing a data-driven alternative: instead of static freezing or depth-based scaling, we use explicit layer-wise alignment signals to dynamically guide update magnitudes during fine-tuning.

## Evaluation

We evaluate our approach along geometric and behavioral dimensions to assess alignment quality and downstream usefulness. For geometric evaluation, we measure layer-wise alignment using Procrustes distance and average cosine similarity between parallel sentence embeddings. We also compute Centered Kernel Alignment (CKA) between pretrained and fine-tuned representations to quantify representational drift, allowing us to assess whether poorly aligned layers improve while well-aligned layers are preserved.

For behavioral evaluation, we measure cross-lingual sentence similarity by comparing cosine similarity between translation pairs and non-parallel pairs. We also test generalization by evaluating on both the trained language pair (English–Spanish) and additional bystander languages from FLORES-200 to determine whether improvements transfer beyond the training distribution.

We compare against three baselines: pretrained XLM-R without fine-tuning, vanilla contrastive fine-tuning with InfoNCE (uniform updates across layers), and a static layer-freezing approach. Performance will be evaluated based on alignment quality and robustness across languages.

## Feasibility and Implementation

We implement our approach using XLM-R base (12 layers), with sentence embeddings obtained via mean pooling and L2 normalization. Training uses contrastive learning with InfoNCE loss and in-batch negatives, optimized with AdamW. Layer-wise alignment scores are computed using average cosine similarity (with optional CKA), and representational drift is controlled via an L2 constraint between pretrained and fine-tuned representations. Updates are scaled by adjusting per-layer learning rates based on alignment scores.

To ensure efficiency, alignment scores are computed on mini-batches and updated periodically rather than every step. We fine-tune on a subset of OPUS-100 English–Spanish bitext and evaluate on FLORES-200, comparing against pretrained XLM-R, vanilla contrastive fine-tuning with InfoNCE (uniform updates), and a static layer-freezing baseline. Initial experiments focus on English–Spanish, with extension to additional languages if time permits.

The system will be implemented in PyTorch using the HuggingFace Transformers library with minimal modifications to standard training loops. Alignment scoring reuses intermediate representations and introduces no additional parameters. Training is feasible on a single GPU (e.g., Google Colab) using mixed precision and moderate batch sizes. Logging will be handled with tools such as Weights & Biases or TensorBoard to monitor alignment trends and training stability. The modular design allows components to be developed and validated independently before integration.